

Group Hands-On Laboratory Examination

Summary

- **Duration:** 3 Hours (180 Minutes)
- **Target Level:** 4th Year BSIT (Beginner in Generative AI Tools)
- **Group Size:** 3–4 Students per Group
- **Deliverable:** One shared GitHub Repository containing the project source code, database scripts, unit tests, and a consolidated SUBMISSION.md report.

Team Member Role Assignments

Before starting, the group must assign specific primary responsibilities to each member:

Member	Assigned Role	Core Responsibilities
Member 1	Systems Architect & Prompt Lead	Task 1 (Requirements & Prompt Engineering) + Task 5 (Documentation & Integration)
Member 2	Frontend Engineer	Task 2 (AI-Assisted UI & WCAG Accessibility)
Member 3	Database & Backend Engineer	Task 3 (3NF Schemas, Mermaid.js ERD & SQL Scripts)
Member 4 (if 4 members)	QA & Security Engineer	Task 4 (Shift-Left Unit Testing & Vulnerability Refactoring)

Note for groups of 3: Member 1 and Member 3 will split the responsibilities of Task 4.

System Scenario / Problem Statement

Your team is tasked with building a working prototype for an **Online Campus Event Management System**. The system must allow students to view upcoming campus events, register for an event, and allow administrators to view registered attendees.

Instructions

Task 1: Requirements Analysis & Prompt Architecture (30 Mins | 20 Points)

Lead: Member 1 (Systems Architect)

1. Construct a **Production-Grade Prompt** using the **Role–Context–Task–Constraints (RCTC)** framework to request an overall system design from an AI model.
 - a. Include a **Persona** (e.g., Lead Systems Architect).
 - b. Include at least one **Negative Constraint** (e.g., "Do not use third-party state management libraries like Redux").
2. Record the **exact prompt** and **AI output** in SUBMISSION.md under ## Task 1.
3. **Manual Grounding Evaluation:** Write a 3–4 sentence evaluation verifying if the AI-generated architecture is realistic for a 3-hour team prototype.

Task 2: AI-Assisted Frontend Development (45 Mins | 25 Points)

Lead: Member 2 (Frontend Engineer)

1. Use an AI tool (or prompt-based UI generator like v0) to build a prototype interface for the **Event Catalog & Registration Form**.
2. **Technical Requirements:**
 - a. Structure the page using **Semantic HTML5** tags (<header>, <main>, <section>, <article>, <footer>) instead of generic container wrappers.
 - b. Implement basic **WCAG Accessibility (POUR)** features: include aria-label attributes for input fields, proper form labels, accessible color contrast, and image alt text.
3. Commit and push all UI code into the /frontend directory of your group repository.

Task 3: Database Design & ERD Generation (45 Mins | 25 Points)

Lead: Member 3 (Database Engineer)

1. Prompt an AI tool to design a **3rd Normal Form (3NF)** schema for the Event Management System covering at least 3 relational entities (e.g., Users, Events, Registrations).
2. **Visual ERD Deliverable:** Instruct the AI to output an Entity-Relationship Diagram in **Mermaid.js** code format. Embed this block into SUBMISSION.md.
3. **Database DDL Script:** Generate a production-grade SQL script.
 - a. Explicitly request Foreign Key rules, CHECK constraints, and **non-clustered indexes on foreign key columns**.
4. Save the script in the group repository as /database/schema.sql.

Task 4: Shift-Left Testing, Security & Refactoring (45 Mins | 20 Points)

Lead: Member 4 (QA Engineer) or Shared between Members 1 & 3

1. **Unit Test Generation:** Prompt an AI assistant to write **Unit Tests** for a core validation routine (e.g., validating student email domains like @univ.edu.ph or verifying seat availability) using **Mock Objects** to isolate external dependencies.
2. **Security & Vulnerability Challenge:** Review and fix the following intentionally flawed backend method:

```
// Flawed code: Contains SQL Injection and unmanaged resource leak
public string GetUserRegistration(string inputEmail) {
    string connStr = "Server=myServerAddress;Database=myDataBase;User
Id=myUsername;Password=myPassword;";
    SqlConnection conn = new SqlConnection(connStr);
    conn.Open(); // Connection is not closed or disposed
    SqlCommand cmd = new SqlCommand("SELECT * FROM Registrations WHERE
Email = '" + inputEmail + "'", conn);
    return cmd.ExecuteScalar().ToString();
}
```
3. Prompt an AI tool to diagnose the code for **SQL injection risks and memory leaks**.

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4. Refactor the snippet using parameterized queries and proper resource disposal (using statements). Save the refactored solution under `/backend/RegistrationService.cs`.

Task 5: Group Integration & Verification Report (15 Mins | 10 Points)

Lead: Member 1 (with input from all members)

Create a consolidated `SUBMISSION.md` in the root folder containing:

1. **Team Roster:** List all group members and their assigned roles.
2. **Setup Instructions:** Steps on how to run and view the project files.
3. **AI Disclosure Statement:** Declaration of all AI tools used during the exam and how outputs were verified.
4. **Group Verification Log Table:** Document at least 3 distinct instances across the project where your team manually corrected or refined AI-generated output to fix errors, security flaws, or bad practices.

Verification Log Format Example:

Task #	Identified AI Flaw / Limitation	Manual Correction Applied	Member Responsible
Task 2	Missing aria-label tags on input fields	Manually added accessible WCAG labels	Member 2
Task 3	AI forgot to add foreign key indexes	Added non-clustered index creation scripts	Member 3
Task 4	AI refactored code without closing DB connection	Added C# using blocks for proper disposal	Member 4

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Grading Rubrics

Criteria	Task 1: Prompting	Task 2: Frontend UI	Task 3: Database	Task 3: Database	Task 5: Documentation	Total
Points	30	20	20	15	15	100

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